

Haeyun Choi

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Research Interests

My research interests broadly lie in computer vision, particularly in approaches that prioritize real-world applicability. I am currently focusing on developing robust 3D scene representations and rendering techniques that remain effective under challenging, non-ideal conditions. My research goal is to move beyond idealized assumptions, embracing the complexity of real-world scenarios.

Education

MS	POSTECH , Artificial Intelligence	Mar 2023 – Feb 2025
	• Advisor : Prof. Sunghyun Cho • Thesis : "<i>Exploiting Deblurring Networks for Radiance Fields</i>"	
BS	Korea University , Biomedical Engineering	Mar 2015 – Aug 2022

Publications

Exploiting Deblurring Networks for Radiance Fields	2025
Haeyun Choi , Heemin Yang, Janghyuk Han, Sunghyun Cho IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2025) Workshop on Image Processing and Image Understanding (IPIU 2025, oral)	
Optical Flow-based Shakiness Detection in Dash-cams for Hit-and-Run Accident Analysis	2024
Haeyun Choi , Sunghyun Cho Workshop on Image Processing and Image Understanding (IPIU 2024)	
AutoCycle-VC: Towards Bottleneck-Independent Zero-Shot Cross-Lingual Voice Conversion	2023
Haeyun Choi , Jio Gim, Yuho Lee, Youngin Kim, Young-Joo Suh arXiv preprint	

Awards and Honors

Outstanding Paper Award, IPIU 2025 (link)	2025
KT-POSTECH AI Expert Program Scholarship \$7,800 x 4 (link)	2023 - 2025
Excellence Prize, POSCO Youth Academy AI Competition	2022
Grand Prize, AI for Health & SDGs Competition (link)	2021
Hongsan Scholarships \$1,700	2021
Mirae Asset Exchange Student Scholarships \$3,600 (link)	2021

Experience

KT R&D Center , AI Research Scientist	Seoul, South Korea
• Visual localization for AI-based navigation in AR glasses	Mar 2025 – Present
POSTECH Institute of Artificial Intelligence , Research Scientist	Pohang, South Korea
• PI : Prof. Young-Joo Suh • Voice conversion using mel-spectrogram representation	May 2022 – Feb 2023

Projects

Real-time Danger Alert for Convenience Stores by Motion and Voice Recognition

2022
poster(soon) 

- Designed a real-time violence detection system for store clerks
- Used top-down pose estimation and lightweight classification models
- Built a 20k image dataset and enabled SMS alerts with incident frames

Wear Your Mask On: Real-time Mask Detection & Warning System

2021
poster(soon) 

- Built a real-time mask alert system for COVID-19, supporting public safety
- Enabled real-time video inference using MobileNet and Haar Cascade

Extracurricular Activities

Team-Mate Mentorship Program (K-Digital Training , KCCI)

Aug 2024 – Present

- Selected as 1st cohort mentor  in a national career development program
- Peer mentorship program, supporting junior trainees in career navigation
- Engaged in job camps  and social contribution programs 

Military Service (Republic of Korea Air Force)

Aug 2016 – Aug 2018

- Sergeant as Operations
- Maintained 24/7 emergency standby status for mission deployment
- Managed mission planning and scheduling for search and rescue operations